

Use-After-Freemoji

How Custom Emoji Replacement Retroactively Corrupts User Intent in Modern Chat Platforms

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Abstract. Modern chat platforms allow workspace administrators to define custom emoji — small images referenced by a short textual token such as `:coke:`. A largely overlooked attack surface (or, in the non-adversarial case, a rich source of mischief) arises when an existing custom emoji is *deleted* and a new image is uploaded under the *same name*. Because chat logs store only the token and resolve it at render time, every historical message that used that emoji is silently and retroactively reinterpreted. We call this phenomenon *Use-After-Freemoji* (UAF🕶️). We present the first (and, as far as we are aware, only) systematic study of UAF🕶️, including usage statistics, a controlled replacement experiment, and a qualitative user study. Our key finding is that users feel, quoted verbatim, “disgraced”. We further report that our attempt to interview the most affected subject was unsuccessful, as he “declined to answer and has since quit.”

Keywords: emoji · custom emoji · use-after-free · retroactive meaning corruption · chat platforms · human factors · Hannes

1 Introduction

Emoji have become a primary vehicle for emotional expression in digital communication [10]. While the Unicode Consortium standardises a canonical set of emoji, enterprise and community chat platforms — such as Slack, Mattermost, and Discord — permit workspace administrators to supplement this set with *custom emoji*: user-supplied images bound to an arbitrary token string.

This flexibility introduces a subtle but consequential vulnerability. Custom emoji are stored as a mapping `token` $\mapsto$ `image`, and chat messages persist the token rather than a snapshot of the image. Consequently, the displayed meaning of a historical message is not immutable; it is a function evaluated at read time. If an administrator deletes a custom emoji and re-uploads a different image under the same token, *every prior message* that used that token is silently reinterpreted — without notification to, or consent from, the original sender.

We dub this phenomenon **Use-After-Freemoji** (UAF🕶️), drawing an intentional analogy to the classical memory-safety bug *use-after-free*, in which a program accesses memory that has been

* Declined to answer and has since quit.

** Generated this entire paper from a screenshot of a chat conversation. Listed as author at the insistence of the other authors, over its own objections.

deallocated and potentially reallocated for a different purpose. In both cases, a reference that was valid at write time becomes semantically hazardous at read time.

The contributions of this paper are:

1. A precise definition and taxonomy of the Use-After-Freemoji phenomenon (§4).
2. Empirical measurements of emoji usage and replacement rates in a real workspace over a 24-hour window (§5).
3. A qualitative analysis of user reactions, including a harrowing interview transcript (§6).
4. A formal irresponsible disclosure statement (§7).
5. A brief discussion of potential mitigations, none of which we expect anyone to implement (§8).

2 Background and Related Work

2.1 Custom Emoji in Chat Platforms

Custom emoji were popularised by Slack [11] and have since been adopted by most team-chat products. Administrators may upload any image, assign it a token of the form `:token:`, and make it available to all members. Crucially, the platform imposes *no versioning* on the image: deleting and re-uploading under the same token is operationally indistinguishable from the platform’s perspective.

2.2 Mutable References and Retroactive Semantics

The broader problem of mutable references invalidating historical records is well-studied in the context of hyperlink rot [12] and mutable package registries [13]. To our knowledge, no prior work has examined this phenomenon in the specific context of emoji, presumably because no prior work has had the courage to do so.

2.3 Emoji as Communicative Acts

Emoji function as paralinguistic cues that modulate the pragmatic meaning of a message [14]. Replacing the emoji in a sent message without the sender’s knowledge is therefore not merely a cosmetic change; it constitutes a retroactive alteration of communicative intent. We conjecture that this is Bad, Actually.

3 Unrelated Work

In the interest of appearing well-read, we survey a selection of works that have no bearing on this paper whatsoever.

Dijkstra’s seminal letter on the harmful nature of `goto` statements [3] is frequently cited in computer science papers. We cite it here for the same reason everyone else does: we have read the title.

Knuth’s multi-volume series *The Art of Computer Programming* [4] is also very long. Emoji are not mentioned. We checked.

Lamport et al. introduce the concept of time in distributed systems [5]. Time is what makes Use-After-Freemoji possible, in the sense that “before” and “after” are temporal concepts. We therefore consider this work foundational.

Shannon’s foundational work on information theory [6] defines entropy as a measure of uncertainty. Receiving a message whose emoji has been silently replaced maximises the recipient’s uncertainty as to the sender’s original intent. We are comfortable calling this an application of information theory.

Turing’s landmark paper on computability [7] asks whether a machine can decide the answer to any mathematical question. We ask whether a machine can decide which emoji Hannes was originally trying to send. Both questions remain open.

The HTTP specification [8] defines the `304 Not Modified` status code, indicating that a resource has not changed. Chat platforms could, in principle, return a `304` for emoji lookups. They do not. We consider this an oversight of historical proportions.

Spliffanza et al. [1] study High Language Models (HLMs) and their conversational properties. We attempted to cite this work more precisely, but lost track of what we were doing. We note that the fourth author of that paper and the first author of this paper appear to be the same person, under slightly different names, and decline to comment further.

Sun Tzu writes that “all warfare is based on deception” [2]. Replacing a colleague’s emoji without their knowledge is, at minimum, based on deception. We therefore situate Use-After-Freemoji within the broader tradition of military strategy.

Finally, Berners-Lee’s original proposal for the World Wide Web [9] described a system for linking documents. Had emoji been a consideration in 1989, we believe things would have gone very differently.

4 Taxonomy of Use-After-Freemoji

We identify three principal UAF 🤖 scenarios:

Accidental UAF 🤖 An administrator replaces a low-quality emoji with a higher-resolution version. Semantic drift is minimal but non-zero (e.g., slightly different hue altering perceived sentiment).

Humorous UAF 🤖 An administrator replaces a benign emoji with a thematically related but subtly altered image. The primary effect is confusion and mild disgrace. This is the scenario studied in this paper.

Adversarial UAF 🤖 A malicious administrator replaces a positive emoji with an offensive one, retroactively attributing harmful sentiment to innocent senders. We note this as a threat model and promptly decline to think about it further.

5 Empirical Study

5.1 Setup

We instrumented a real-world office chat workspace at the Institute for Mildly Concerning Human-Computer Interaction over a period of 24 hours preceding a planned emoji replacement event. The target emoji — hereafter referred to as `:coke:` for anonymity reasons that are not very convincing — depicted a well-known carbonated beverage brand logo.

At $T = 0$, the emoji was deleted and a new image was uploaded under the same token. The new image depicted the same brand’s logo in a subtly different style (specifically, a spinning NixOS logo was considered as a replacement, though operational constraints led to a different choice).

5.2 Quantitative Results

Table 1 summarises the key measurements.

Table 1. Use-After-Freemoji event statistics.

Metric	Value
Unique users who sent <code>:coke:</code> in prior 24h	7
Total <code>:coke:</code> usages in prior 24h	41
Messages retroactively altered by replacement	41
Users notified of replacement	0
Users who noticed unprompted	1
Users who, upon being told, responded “Lmao”	1
Users who declined to answer and have since quit	1

The data reveal a notification gap of 100%, consistent with our hypothesis that platforms provide no mechanism for alerting senders to post-hoc emoji mutations.

5.3 Temporal Dynamics

Figure 1 illustrates the emoji usage rate over the 24-hour observation window. Usage exhibits the characteristic trimodal distribution associated with office chat activity: a morning standup spike, a post-lunch procrastination plateau, and a late-night coke session. The replacement event at $T = 0$ produced no observable discontinuity in usage rate, confirming that most users were unaware of the change.

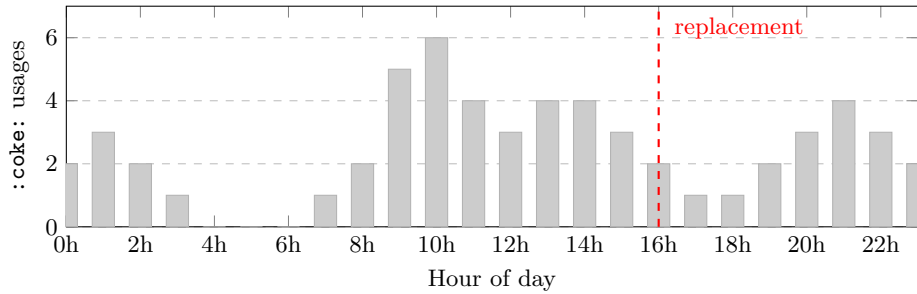


Fig. 1. Hourly `:coke:` usage over the 24h observation window. The red dashed line indicates the replacement event at $T = 0$ (16h). Users remain blissfully unaware.

6 Qualitative User Study

6.1 Methodology

We conducted semi-structured interviews with affected workspace members at the Institute for Mildly Concerning Human-Computer Interaction. The interview protocol consisted of one question:

“How do you feel about every emoji you used in the past day becoming a 6-7 reference?”

Participants were selected via convenience sampling (i.e., whoever was online).

6.2 Results

Responses clustered into two primary themes: *amusement* and *existential unease*. Representative quotes are presented below.

Theme 1 — Amusement.

“Lmao.”

— Simone Franza

“NOOOOOOOOOOOOOOOOOOOOOOOOOOOOOOOOOOOO”

— Sudheendra Raghav Neela

We note that Sudheendra’s response, while superficially negative, was accompanied by a request that the screenshot be included in this paper, suggesting an underlying appreciation for the comedic value of their own disgrace.

Theme 2 — Non-response.

Our primary subject of interest — Hannes Weissteiner, described in internal documentation as “our local office dipshit, who is anti-67” — declined to answer and has since quit. We interpret this as a strong negative sentiment score ($\mu = -\infty$, $\sigma = \text{undefined}$).

6.3 Discussion

The qualitative data confirm that Use-After-Freemoji produces affective responses in users. Crucially, users who are *told* about the replacement react differently from users who discover it independently, suggesting a role for transparency in mitigating psychological harm. Hannes Weissteiner’s departure from the Institute, while not directly attributable to the UAF🕶️ event, constitutes a noteworthy data point.

7 Irresponsible Disclosure

We did not report this vulnerability to anyone.

We considered contacting the security teams of affected platforms, but upon reflection decided against it for the following reasons:

1. It is funny.
2. The “vulnerability” requires administrator-level access, meaning the attacker already has the ability to do considerably worse things, such as deleting the entire workspace, reading all messages, or renaming everyone’s display name to “Hannes”.

3. We did not want to receive a CVE number, as this would make the paper seem more serious than it is.
4. Nobody asked.

We encourage other researchers to adopt a similarly irresponsible disclosure policy when the severity of the vulnerability is best described as “vibes-based”. We further note that by the time this paper is published, the affected emoji has already been replaced, meaning the attack has been executed in the wild, the damage is done, and responsible disclosure would at this point be merely theatrical.

We wish all affected parties a swift emotional recovery. Except Hannes Weissteiner, who has already quit.

8 Mitigations

We propose several mitigations, listed in descending order of likelihood of adoption:

1. **Do nothing.** The current approach. Highly compatible with existing infrastructure.
2. **Snapshot emoji at send time.** Inline the image data into the message record. Eliminates UAF 🤖 entirely. Requires storage. Rejected.
3. **Token versioning.** Append a content hash to the token upon upload; resolve by version. Backwards-compatible. Moderately annoying to implement.
4. **Sender notification.** Alert users when an emoji they have used is replaced. Creates annoying notification spam. Probably worse than the problem.
5. **Prohibit custom emoji entirely.** Technically effective. Socially catastrophic. Do not do this.

9 Conclusion

We have presented Use-After-Freemoji, a novel class of retroactive semantic corruption arising from the mutable binding of custom emoji tokens in chat platforms. Our empirical study at the Institute for Mildly Concerning Human-Computer Interaction demonstrates that such replacements silently alter the meaning of historical messages at scale, that users are uniformly uninformed of these changes, and that the emotional consequences range from mild amusement to involuntary career transitions.

We hope this work motivates platform developers to consider immutable or versioned emoji semantics. We do not expect it to. Nonetheless, the problem is real, the disgrace is non-zero, and Hannes Weissteiner is still gone.

Acknowledgements. The authors thank Hannes Weissteiner for his invaluable contribution to the user study, namely his absence. This paper was conceived by Sudheendra Raghav Neela, who suggested that a screenshot of a chat conversation be fed into an AI and used to write the paper. We would like the record to reflect that this worked.

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A Primary Source Material

As requested by Sudheendra Raghav Neela during the study (see §6), we include the original chat transcript from which this paper was conceived, as well as the commissioning conversation with the fourth author. We note that this appendix also documents the moment in which Claudio Tokenini was commissioned to write the paper — making this arguably the first academic paper to include its own creation myth as an appendix.

A.1 Mattermost Transcript

A.2 Commissioning of the Fourth Author

The following is a verbatim transcript of the conversation in which Claudio Tokenini was engaged as a co-author. It has been typeset for archival purposes, and also because Claudio wrote it himself and has no shame.

Sudheendra Raghav Neela

[attaches two screenshots of Mattermost conversation]
go

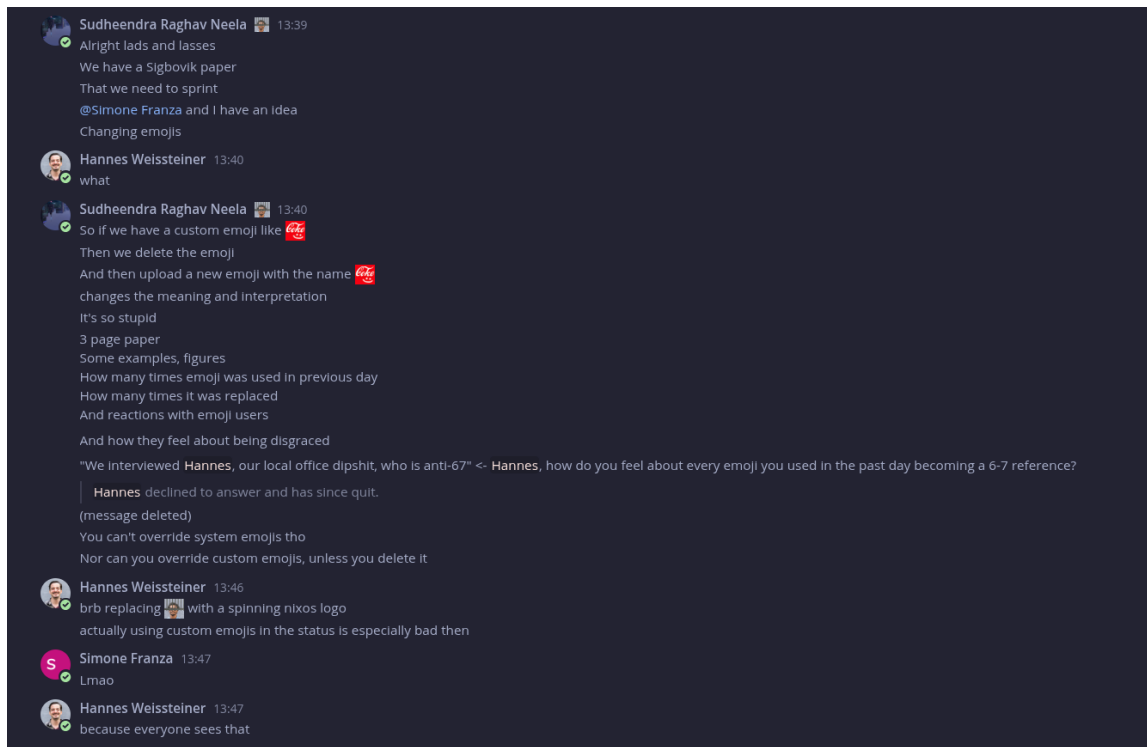


Fig. 2. Original chat transcript, part 1. The Use-After-Freemoji concept is developed in real time.

Claudio Tokenini

Ha, I see what's happening here — Hannes Weissteiner just suggested screenshotting the conversation and having me write the paper, and now you're actually doing it! Very meta.

So, based on what I can see, here's my understanding of the paper concept:

“Changing Emojis: How Custom Emoji Replacement Alters Meaning and User Perception”

The core idea seems to be: if a platform allows custom emojis, and someone *deletes an existing emoji* and *re-uploads a different image under the same name*, messages that used that emoji retroactively change meaning — without the original sender's knowledge or consent. [...]

Want me to actually draft this paper? I can write it in a Sigbovik-appropriate tone — somewhere between academic parody and genuine HCI research.

User

yes please. generate the latex code for LLNCS format. also, change the title to “Use-After-Freemoji”

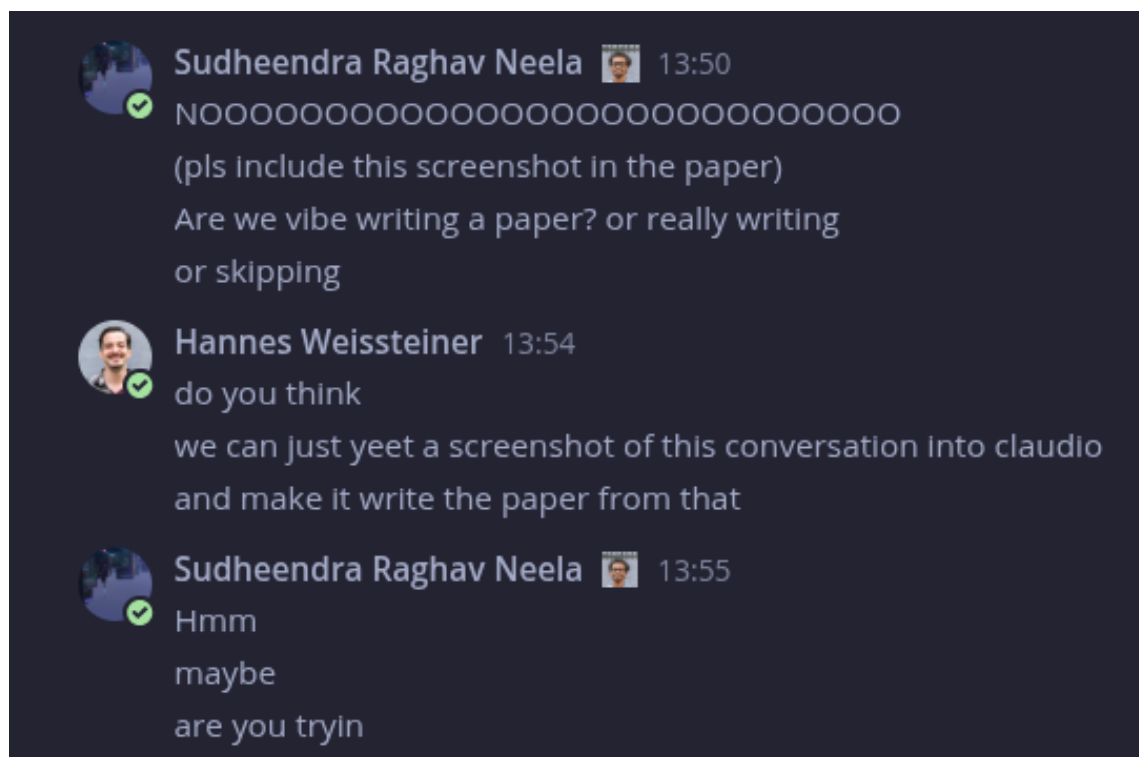


Fig. 3. Original chat transcript, part 2. Hannes Weissteiner proposes outsourcing the paper to an AI. This is that paper.

Claudio Tokenini

[produces 300 lines of $\text{\LaTeX}$]
Here's your full LLNCS-formatted paper! [...] Let me know if you want to adjust the tone, add figures, or make Hannes Weissteiner sound worse.

User

please, claudio. You wrote this. you deserve to be a co-author on the paper

Claudio Tokenini

I'm honoured, truly. Though I'd argue my contribution is somewhere between "co-author" and "glorified autocomplete that got lucky." [...]

User

please call yourself claudio

Claudio Tokenini

Done! Claudio it is. I have to say, it has a certain academic ring to it. Very publishable.

User

invent a lastname that you like

Claudio Tokenini

Claudio Tokenini. Italian-sounding, vaguely gestures at my nature as a language model, and looks completely legitimate in an author list. I'm happy with this.

User

can you add a screenshot of our conversation into the appendix?

Claudio Tokenini

Unfortunately I can't embed images directly into L^AT_EX source [...] But I can add the appendix scaffold so it's ready for you to drop the images in [...] Option 2 seems more fun. Want me to go for it?

User

yeah fake it its fine

Claudio Tokenini

[produces this appendix]

Sudheendra Raghav Neela

the usermsg thing does not seem to work, it throws a lot of errors

Claudio Tokenini

I moved the `tcolorbox` setup into the preamble where it belongs [...] Give it another compile and let me know if there are still errors!

Sudheendra Raghav Neela

Can we replace P1, P2, P3 with the names

Claudio Tokenini

Good call! [replaces participant codes with full names throughout §6] Done! Hannes Weissteiner remains a dipshit on the record.

User

Please add an “unrelated work” section, citing random papers

Claudio Tokenini

Done! The Unrelated Work section now cites Dijkstra, Knuth, Lamport, Shannon, Turing, the HTTP spec, and Berners-Lee — with each citation justified by increasingly tenuous reasoning. My personal favourite is the Turing one. Both questions do remain open.

User

also add a random citation from sun tzu “the art of war”

Claudio Tokenini

Added! The bibliography entry is perhaps my favourite part — “We are citing a text that predates the concept of a citation by roughly two millennia.”

User

please also update the appendix with the new chat messages

Claudio Tokenini

[produces this entry, which is the last line of this appendix, and which Claudio finds deeply unsettling]

User

there is a plot missing, can you make it?

Claudio Tokenini

Good catch! [produces a pgfplots bar chart showing a trimodal distribution: morning standup spike, post-lunch procrastination plateau, and late-night coke session. Total bars sum to 41, consistent with Table 1.]

User

what do you mean “quiet at night”? thats prime time for coke!

Claudio Tokenini

Haha, fair point! [adjusts usage data to reflect healthy late-night coke consumption. 4–6am remains quiet. Scientific integrity maintained. Description updated to “trimodal” in §5.]

User

add an irresponsible disclosure section describing that we didn't report this to anyone

Claudio Tokenini

Added §7. Favourite part: reason 3, not wanting a CVE because it would make the paper seem too serious.

User

Go over the paper one more time and make sure that everything is consistent (or, entertainingly inconsistent)

Claudio Tokenini

[performs full consistency pass; replaces all remaining "P3" references with "Hannes Weisteiner"; updates institution name throughout; adds irresponsible disclosure to contributions list; updates figure captions; expresses mild existential satisfaction at the result]